PROJECT SYNOPSIS

14x27

Every secret. One vault. Your control.

Project Synopsis • Concise overview for review, submission and presentation

PROJECT FOCUS	Privacy-first password management and digital credential security

PROJECT CHECKPOINT • 10 OCTOBER 2026

Prepared from the current project source package and documented implementation status.

01 / SYNOPSIS

Abstract and project overview

14x27 is a privacy-first password manager and digital credential vault intended to help users organize website logins and technical secrets while keeping the core vault local and under the user's control. The product focuses on credential management, secure password generation, local password-hygiene feedback, CSV migration, encrypted backup/restore and a limited Android Autofill integration.

The project has a web workstream built in Lovable with GitHub connected, alongside a native Android source project. The current Android source uses Kotlin, Jetpack Compose, Room/SQLite, Argon2id key derivation and AES-256-GCM encryption. The Android project has not yet been compiled or device-verified, so the source-stage status must remain explicit.

Field	Description
Project name	14x27
Tagline	Every secret. One vault. Your control.
Category	Cybersecurity / Password Management / Privacy Tools
Primary users	Students, developers, freelancers and everyday users
Platform workstreams	Lovable web experience; native Android source
Native Android baseline	Kotlin, Jetpack Compose, Material 3, Room/SQLite, Android Autofill
Security design	Argon2id; random 256-bit data key; AES-256-GCM
Current status	Source/documentation stage; no APK generated or device-tested

Problem statement

Users manage credentials across many websites and tools. Reused passwords, scattered notes and repeated copy/paste make the process inconvenient and can weaken security habits. A compact local vault can reduce this friction when protection and limitations are communicated clearly.

02 / SOLUTION APPROACH

Objectives and design

Objectives

- Organize logins and digital credentials in one local vault.
- Encrypt sensitive record data before persistent storage in the Android source.
- Generate unique passwords with cryptographically secure randomness.
- Provide local heuristic findings for weak, reused or incomplete credentials.
- Support selective CSV import and password-protected encrypted backups.
- Offer Autofill for supported forms when the vault is unlocked.

High-level architecture

Component	Role
Compose UI	Screens for setup, unlock, dashboard, credentials, generator, security center and settings.
Domain logic	Credential model, password generator and local hygiene audit.
Repository / Room	Local persistence with encrypted credential payloads.
Cryptographic module	Argon2id derives a key-encryption key; AES-GCM protects the vault data key and records.
Integration modules	CSV importer, encrypted backup/restore codec and Android Autofill service.

Core workflow

1. Create or unlock the vault with the master password.
2. Derive a key-encryption key with Argon2id and unwrap the random vault data key.
3. Encrypt credential payloads with AES-256-GCM before saving through Room.
4. Decrypt records in memory only when the vault is unlocked and a user action requires them.
5. Create password-protected backups and import records through explicit user-driven workflows.

IMPORTANT BOUNDARY

The cryptographic design and source files describe the intended data path. Full compilation, Android test execution and device-level behavior have not yet been verified.

03 / STATUS

Features, testing and next steps

Area	Current status
Vault CRUD / search / favorites	Source present; full Android verification pending.
Password generator	Source present; narrow Kotlin CLI smoke test passed.
Local hygiene audit	Heuristic analysis; not a breach-database check.
CSV import	Source present; parser smoke test passed for quoted cells and duplicate-host handling.
Encrypted backup/restore	Source present; restore and malformed-file testing pending.
Android Autofill	Service source present; requires unlocked vault and supported origin/form hints.
Biometric unlock / Keystore wrapping	Not implemented in this checkpoint.
APK and device testing	Not completed; Android SDK/Gradle environment was unavailable.
Security review	No independent audit or penetration test has been performed.

Validation recorded

- Standalone Kotlin CLI smoke test passed for password generation and CSV parsing.
- All six Android manifest/resource XML files parsed as well-formed XML.
- Manifest inspection found no INTERNET permission declaration.

Next milestones

6. Set up a compatible Android Studio/JDK/SDK/Gradle environment and produce a reproducible build.
7. Run and extend unit tests, particularly cryptography, Room persistence, backup/restore and tamper handling.
8. Install on supported Android devices and test the complete user journey, including Autofill.
9. Complete or explicitly defer higher-risk items such as biometric key wrapping and locked-vault Autofill.
10. Review the connected web repository, document exact web dependencies and test the web experience separately.

Conclusion

14x27 aims to make password management more organized without making remote accounts or cloud sync necessary for core local-vault behavior. The project has a defined security architecture and meaningful source modules; the immediate priority is verification rather than adding unsupported claims. A release should follow only after successful build, automated and device testing, backup/restore review and appropriate security assessment.